



Fluid Flow in Pipe Systems

Group 3

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Chemical Engineering Lab 1

9/19/19

Introduction



There are 4 main goals

- Verify the conversion factor for the rotameter
- Experimentally calculate a value for the orifice flow coefficient
- Experimentally calculate friction factors as a function of Reynolds number
- Experimentally calculate the loss coefficients for pipe fittings



Theory

For each problem in the experiment, a set of pressure and volumetric flow measurements are made, from which constants are derived.

Orifice Meter: the flow through an orifice and the pressure drop across it are related by the equation


$$Q = C_f A_o \sqrt{\frac{2\Delta P}{\rho}}$$

Pipe Friction Head Loss: the drop in pressure across the pipe and the friction factor are related by the equation

$$h_L = \frac{\Delta P}{\rho} = f \frac{L}{D} \frac{V^2}{2}$$

Fitting Head Loss: the pressure drop across a fitting, the characteristic length, and the loss coefficient are related by the equations

$$h_L = K \frac{V^2}{2} \quad K = f \frac{L}{D}$$

Example calculation: Smallest plastic Pipe Flow Rate

Rotameter Calibration for C_r rotameter correction factor:


$$C_r = \frac{\frac{V}{t}}{Q_r}$$

$$C_{rot} = \frac{1.00L/1.0425 \text{ min}}{10 \text{ L/min}} = 0.8571$$

$$v = \frac{\text{volume flow rate}}{\text{area of the pipes}} = \frac{V}{\pi r^2}$$

$$v = \frac{18 \frac{L}{min}}{\pi \left(\frac{0.01021}{2} \right)^2} \left(\frac{min}{60 \text{ s}} \right) = 3.66 \frac{m}{s}$$

Reynold's Number:

$$Re = \frac{d u \rho}{\mu}$$
$$Re = \frac{0.01021m * 997 \frac{kg}{m^3} * 3.66 \frac{m}{s}}{0.0001 Pa \cdot s} = 37256$$

Flow Coefficient for Orifice Meter:

$$c_f = \frac{Q}{A_0 \sqrt{\frac{2\Delta p}{\rho}}}$$
$$C_f = \frac{0.000614m^3/s}{.0001158m^2 * \sqrt{\frac{2(32129 Pa)}{997 \frac{kg}{m^3}}}} = 0.66$$

Head Loss in the Pipe h_L for the expansion:

$$h_L = \frac{f l v^2}{2dg}$$

$$h_L = 0.006937 * \frac{1.2m}{0.01021m} * \frac{(3.66\frac{m}{s})^2}{2*9.81\frac{m}{s^2}} = 0.25 \text{ m}$$

Loss coefficient (180° Bend 16.6% Flow Rate):

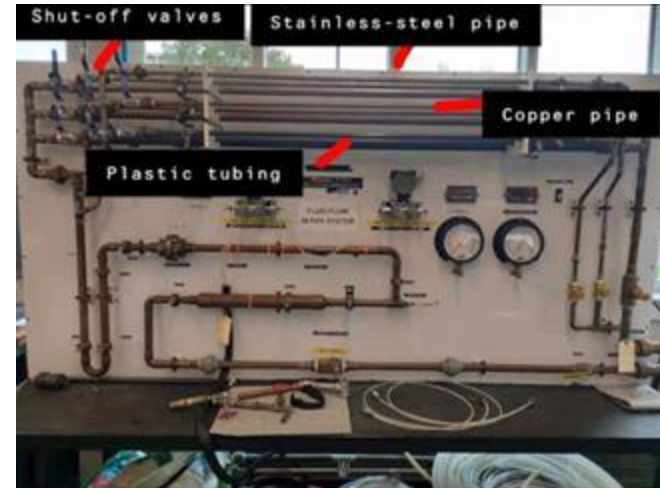
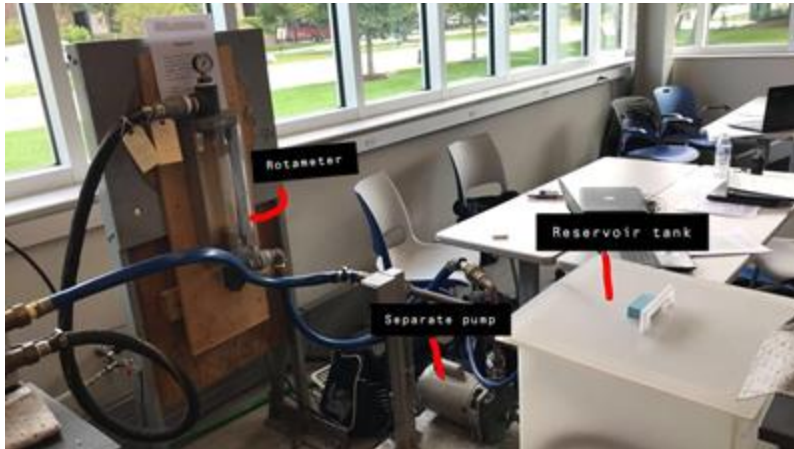
$$K = f \cdot \frac{l}{d}$$

$$K = \frac{0.006937*0.15m}{0.0121m} = 0.0856$$

Apparatus



The pipe friction demonstrator is a reservoir and pump connected to a series of pipe fittings and 9 different parallel pipes that can be individually opened by controlling valves. There is also a differential pressure transmitter (DPM) that is used to measure the pressure change across a pipe or fitting and a rotameter that is used to measure volumetric flow.





To start the demonstrator, fill the reservoir from the hose to the wall until it covers the blow off pipe, open all of the parallel test pipes, switch the pump on, open and close the blow off valve, and wait for bubbles to stop flowing through the rotameter.

To use the DPT, connect it to the connection ports on either side of the fitting then open and close the black tabs on either side of the DPT to open the flow through the tubes.

Material & Supplies



- All materials used in the lab were at the apparatus. Such as pipes and hoses.
- Hampden H-6920 Pipe Friction Demonstrator 
- Differential pressure transmitter connection hoses
- 1L graduated cylinder
- Water
- Rotameter
- Reservoir
- Pump

Procedure



Fill the reservoir, and start the flow by turning on the pump and opening and closing the blow off valve.

Open each pipe individually and measure the flow and pressure drop across it while all other parallel pipes are closed.

For each pipe fitting and the orifice, measure the pressure drop across the fitting and flow rate through it at 3 different flow rates.

For each pipe and fitting record a flow rate and a pressure differential.

Experimental Challenges

The rotameter is not a very accurate way to measure flow, and so there was no noticeable difference in flow between the small and medium pipe sizes.

Pipes leaked

Bubbles flowing through the system can cause inaccuracies in pressure measurements. If bubbles are present, the system should be run for several more minutes to push them out.



Safety

- The only chemical present was water, therefore, gloves and goggles should be worn at all times.

Comment Summary

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5. What correlation was used?
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9. Why is the procedure here?

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10. Some of them are part of the apparatus and not supplies
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